

Arithmetic Power Geometry as an Interpretable Out-of-Distribution Detector for Materials Discovery: A Pilot Study on Matbench Crystal Benchmarks

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Abstract

Machine-learning models for materials discovery are increasingly accurate on standard property-prediction benchmarks, but their reliability may degrade when candidate materials move outside the structural and chemical regimes represented in training data. This study introduces a pilot Arithmetic Power Geometry (APG) framework for interpretable out-of-distribution (OOD) screening in materials discovery. Given a positive material descriptor vector, APG maps squared coordinate mass into a probability distribution and computes normalized entropy, concentration, and exponent-deformation defects. The method is motivated by prior APG Shannon–Rényi control laws and by empirical APG anomaly studies in which entropy collapse and concentration growth are treated as representation-dependent structural signatures rather than universal laws. We evaluate APG descriptors on three Matbench datasets: `matbench_mp_gap`, `matbench_mp_e_form`, and `matbench_perovskites`. A pilot sample of 1,000 materials per dataset is analyzed using composition-derived and structure-augmented APG descriptors. Strong structural separation is observed across the three datasets. When `matbench_mp_e_form` is treated as the in-distribution reference, the APG Mahalanobis one-vs-rest OOD AUROC is 0.8426; the perovskite reference yields an AUROC of 0.9210, reflecting stronger descriptor separability from the two Materials Project tasks. These results should not be interpreted as state-of-the-art materials prediction. Instead, they support APG as a compact, interpretable structural monitoring layer that can be attached to Matbench, Matbench Discovery, graph neural networks, and high-throughput materials screening pipelines.

Keywords: Arithmetic Power Geometry; out-of-distribution detection; materials discovery; Matbench; entropy; concentration; crystal graph learning; interpretable AI.

1 Introduction

Materials discovery increasingly relies on machine learning to accelerate property prediction, stability screening, and high-throughput candidate selection. Benchmarks such as Matbench provide standardized tasks for evaluating materials property predictors across experimental

and computed datasets (Dunn et al., 2020). More recent discovery-oriented benchmarks, including Matbench Discovery, focus on the realistic problem of ranking candidate inorganic crystals by stability and using machine-learning models as pre-filters before expensive first-principles calculations (Riebesell et al., 2025). At the model level, graph neural networks and universal interatomic potentials have produced rapid progress in property prediction, relaxation, and screening (Xie and Grossman, 2018; Chen et al., 2019; Chen and Ong, 2022; Deng et al., 2023). However, high benchmark performance does not automatically imply trustworthy extrapolation to new chemistries, rare structural symmetries, or candidate spaces generated by different procedures.

Out-of-distribution generalization has therefore become a central problem in materials informatics. Recent work argues that many heuristic OOD splits in materials science measure interpolation rather than genuine extrapolation and may overestimate generalizability (Li et al., 2025). Other studies emphasize cross-domain data bias and the need to detect whether the training domain and deployment domain differ in chemically meaningful ways (Trezza and Chiavazzo, 2025). This concern is aligned with broader machine-learning evidence that predictive confidence, uncertainty estimates, and learned representations can fail under distribution shift (Hendrycks and Gimpel, 2017; Lakshminarayanan et al., 2017; Ovadia et al., 2019; Ren et al., 2019; Yang et al., 2024).

Arithmetic Power Geometry (APG) offers a complementary perspective. Instead of asking only whether a model predicts a target property accurately, APG asks how the squared coordinate mass of a material representation is organized across descriptor dimensions. Prior APG work introduced exponent-deformation geometry relative to the Euclidean target and showed that local closure defects are governed by entropy-like quantities (Akhtar, 2026a). A Shannon–Rényi APG control theorem then converted APG defects into normalized power-sum defects bounded by Shannon and Rényi structure (Akhtar, 2026b). A finite-dimensional extension showed that multi-variable APG defects act as bounded deformation descriptors for high-dimensional information systems (Akhtar, 2026c). Applied APG fraud work further proposed entropy collapse and concentration growth as empirical, representation-dependent signatures of anomalous behavior rather than universal theorems (Akhtar, 2026d). The present manuscript transfers this idea to materials discovery: a candidate material may be considered structurally unfamiliar when its APG entropy, concentration, and defect pattern moves outside the learned reference regime.

2 Research Question and Contribution

This study investigates whether APG descriptors can serve as interpretable out-of-distribution indicators for materials discovery. The objective is not to replace graph neural networks, formation-energy predictors, density functional theory, uncertainty-estimation methods, or Matbench benchmark approaches. Instead, the objective is to determine whether APG descriptors reveal measurable structural regimes across Matbench datasets and whether these regimes can support a simple out-of-distribution scoring framework.

This study makes four contributions. First, it formulates APG-OOD descriptors for materials vectors using the entropy, concentration, and defect definitions developed in earlier APG work. Second, it evaluates these descriptors on three Matbench benchmark datasets

representing band-gap prediction, formation-energy prediction, and perovskite materials. Third, it reports a pilot one-versus-rest APG-OOD evaluation using Mahalanobis distance in APG descriptor space. Fourth, it places the APG framework within the broader context of materials benchmarks, graph-neural architectures, uncertainty estimation, and out-of-distribution detection while maintaining conservative and interpretable claims.

3 Related Work

3.1 Materials benchmarks and discovery

The Matbench test set introduced a curated group of materials property-prediction tasks and the Automatminer reference algorithm, enabling reproducible comparison across methods (Dunn et al., 2020). This benchmark structure is important because materials datasets differ strongly in size, target type, input representation, and required inductive bias. Matbench Discovery extends the benchmark philosophy toward stability-oriented discovery, comparing machine-learning energy models in a setting designed to simulate high-throughput searches for stable inorganic crystals (Riebesell et al., 2025). The present work uses Matbench-style datasets not to compete on leaderboard property error, but to evaluate whether APG descriptors reveal cross-task structural differences.

3.2 Graph learning and foundation models for materials

Crystal Graph Convolutional Neural Networks represented crystals directly as graphs and achieved strong property-prediction performance across diverse structures (Xie and Grossman, 2018). MEGNet generalized graph networks for molecules and crystals and became a major baseline for materials property prediction (Chen et al., 2019). ALIGNN introduced atomistic line-graph neural networks that incorporate bond-angle information for improved materials property predictions (Choudhary and DeCost, 2021). M3GNet and CHGNet further moved toward universal interatomic potentials and pretrained atomistic modeling across broad chemical space (Chen and Ong, 2022; Deng et al., 2023). These models primarily optimize predictive performance and physical modeling. APG is proposed here as an auxiliary structural descriptor layer that can be computed on raw descriptors, engineered descriptors, or learned embeddings.

3.3 OOD detection and uncertainty

Classical OOD detection in deep learning includes baseline softmax confidence methods, maximum softmax probability, energy scores, Mahalanobis feature-space scoring, deep ensembles, and calibration-aware uncertainty estimation (Hendrycks and Gimpel, 2017; Liang et al., 2018; Lee et al., 2018; Lakshminarayanan et al., 2017; Liu et al., 2020; Guo et al., 2017). Distribution shift benchmarks and surveys show that OOD evaluation is sensitive to the chosen split, feature representation, and metric (Ovadia et al., 2019; Yang et al., 2024). In materials science, the same concern is amplified because chemical space is highly structured and candidate-generation pipelines may produce distributions unlike training databases (Li

et al., 2025; Trezza and Chiavazzo, 2025). APG does not solve OOD detection generally; it contributes an interpretable, low-dimensional map of representation organization.

Classical information theory provides the mathematical language behind the APG map: Shannon entropy, Rényi entropy, Rényi divergence, majorization theory, and information geometry establish the probability-distribution background against which APG deformation descriptors are interpreted (Shannon, 1948; Rényi, 1961; Cover and Thomas, 2006; Csiszár and Körner, 2011; van Erven and Harremoës, 2014; Marshall et al., 2011; Amari, 2016). Classical inequality and concentration measures, including the Gini tradition, remain important comparison points because APG should be read as an exponent-deformation view of distributional organization rather than as a replacement for established concentration statistics (Gini, 1912). General deep-learning representation work and ImageNet-style benchmark culture also motivate the need for standardized evaluation rather than isolated demonstrations (Krizhevsky et al., 2012; Goodfellow et al., 2016). Additional materials graph-learning studies and surveys show that representation quality, geometric information, and message-passing design remain central to materials ML performance (Cheng et al., 2021; Reiser et al., 2022). Recent OOD methods such as activation rectification further illustrate that internal representation geometry can be useful for distribution-shift detection (Sun et al., 2021).

3.4 APG background

APG studies algebraic closure relations under exponent deformation while fixing positive base coordinates. Its Euclidean baseline principle treats the exponent 2 as the reference state against which deformation is measured (Akhtar, 2026a). The Shannon–Rényi control manuscript established that the local APG closure defect can be represented as a normalized power-sum defect and bounded through entropy structure (Akhtar, 2026b). The multi-variable APG control manuscript extended this construction to finite-dimensional vectors and emphasized that APG descriptors should complement, not replace, established information and concentration measures (Akhtar, 2026c). APG-Fraud then applied entropy, concentration, and defect descriptors to anomaly analysis and explicitly framed entropy collapse and concentration growth as empirical, representation-dependent hypotheses (Akhtar, 2026d). The present manuscript directly relies only on these APG works because they provide the definitions, bounds, multi-variable extension, and applied anomaly framing required for APG-OOD materials discovery.

4 APG-OOD Theory for Materials Vectors

Let a material representation be a positive vector

$$X = (x_1, x_2, \dots, x_m) \in \mathbb{R}_+^m. \quad (1)$$

The vector may contain elemental features, composition-derived descriptors, structure descriptors, or latent neural embeddings. APG converts X into squared Euclidean weights

$$w_i = \frac{x_i^2}{\sum_{j=1}^m x_j^2}, \quad i = 1, \dots, m. \quad (2)$$

Then $W = (w_1, \dots, w_m)$ is a probability vector. The APG entropy is

$$H(W) = - \sum_{i=1}^m w_i \log w_i, \quad (3)$$

and the normalized APG entropy is

$$H_N(W) = \frac{H(W)}{\log m}, \quad m > 1. \quad (4)$$

The APG concentration is

$$C(W) = \max_{1 \leq i \leq m} w_i. \quad (5)$$

For $t \geq 2$, the multi-variable APG defect is

$$D_m(t) = 1 - \sum_{i=1}^m w_i^{t/2}. \quad (6)$$

This is the normalized power-sum form of the Euclidean-target closure defect. Equivalently, for positive coordinates,

$$\delta_K(t) = \left| 1 - \frac{\sum_{i=1}^m x_i^t}{(\sum_{i=1}^m x_i^2)^{t/2}} \right| = D_m(t), \quad t \geq 2. \quad (7)$$

The Shannon–Rényi APG control inequality used as theoretical motivation is

$$D_m(t) \leq 1 - \exp\left(-\frac{t-2}{2}H(W)\right) \leq \frac{t-2}{2}H(W). \quad (8)$$

If $M = \max_i w_i$, the complementary maximum-weight lower bound is

$$D_m(t) \geq 1 - M^{(t-2)/2}. \quad (9)$$

For materials OOD detection, the interpretation is direct. High $H_N(W)$ means squared descriptor mass is spread across many features. High $C(W)$ means one feature dominates. High $D_m(t)$ indicates that the representation changes strongly when the Euclidean exponent is deformed. Therefore, a candidate material can be flagged as structurally unfamiliar when its APG descriptor vector

$$\Phi_{\text{APG}}(X) = (H_N(W), C(W), D_m(3), D_m(4), D_m(5)) \quad (10)$$

falls outside the reference APG distribution learned from in-distribution materials. The orders $t=3,4,5$ were selected to capture deformation behavior across multiple exponent scales while remaining close to the Euclidean reference exponent $t=2$.

5 Methodology

5.1 Datasets

The pilot uses three Matbench datasets: `matbench_mp_gap`, `matbench_mp_e_form`, and `matbench_perovskites`. The first two are Materials Project crystal-structure tasks for band gap and formation energy, while the third contains perovskite structures; the Materials Project itself is a major infrastructure for accelerated materials innovation (Jain et al., 2013). The manuscript uses a pilot sample of 1,000 materials from each dataset. This sampling design is sufficient for a structural feasibility scan, but not for a final leaderboard-quality benchmark.

5.2 Feature construction

For each material structure, we extracted simple composition and structure descriptors. The composition vector used element-weighted atomic-number contributions of the form $(Z_e + 1)n_e/n$, where Z_e is atomic number, n_e is the element occupancy count, and n is the total occupancy count. Structure-augmented descriptors included lattice lengths, lattice angles, volume, number of sites, number of elements, mean atomic number, and atomic-number span. Two APG maps were then computed: a composition-only APG map and a structure-augmented APG map.

5.3 Pilot OOD scoring

For one-vs-rest OOD evaluation, each dataset was treated as an in-distribution reference in turn. APG descriptor vectors were standardized using the reference dataset and scored using Mahalanobis distance in APG descriptor space:

$$S_{\text{OOD}}(z) = (z - \mu)^\top \Sigma^{-1} (z - \mu), \quad (11)$$

where $z = \Phi_{\text{APG}}(X)$, and μ and Σ are the empirical mean and covariance of the in-distribution APG descriptors. Higher scores indicate greater APG structural deviation. AUROC and average precision were computed for distinguishing the reference dataset from the two remaining datasets.

6 Results

Table 1 reports the pilot descriptor summary. The composition-only descriptors indicate that `matbench_mp_e_form` exhibits the lowest mean normalized APG entropy and the highest mean concentration among the three datasets. The perovskite dataset has the highest mean D_4 , indicating stronger exponent-deformation response under the chosen feature map.

Table 1: APG descriptor summary for the Matbench pilot sample.

Dataset	Descriptor	Value
mp_e_form	n	1000
mp_e_form	Target mean	1.9122
mp_e_form	Target SD	0.3236
mp_e_form	H_N	0.5926
mp_e_form	C	0.7175
mp_e_form	D_4	0.3751
mp_e_form	Struct. H_N	0.2195
mp_e_form	Struct. C	0.8330
mp_e_form	Struct. D_4	0.2466
mp_gap	n	1000
mp_gap	Target mean	0.6595
mp_gap	Target SD	1.1867
mp_gap	H_N	0.6264
mp_gap	C	0.6857
mp_gap	D_4	0.4187
mp_gap	Struct. H_N	0.4857
mp_gap	Struct. C	0.5918
mp_gap	Struct. D_4	0.5501
perovskites	n	1000
perovskites	Target mean	1.4732
perovskites	Target SD	0.7563
perovskites	H_N	0.6179
perovskites	C	0.6646
perovskites	D_4	0.4551
perovskites	Struct. H_N	0.4048
perovskites	Struct. C	0.6974
perovskites	Struct. D_4	0.4544

Target mean and target SD refer to the benchmark target variable associated with each dataset.

Figure 1 summarizes the mean APG entropy, concentration, and fourth-order defect across the three Matbench pilot datasets and highlights systematic differences in descriptor organization.

To examine distributional behavior beyond mean values, Figure 2 presents the distribution of normalized APG entropy across the three datasets.

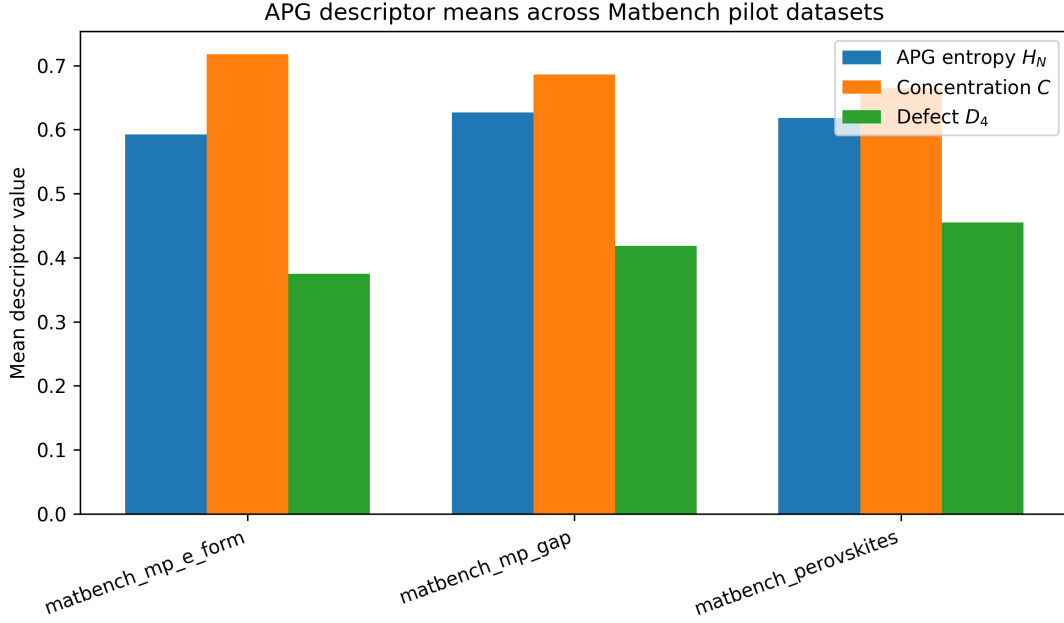


Figure 1: Mean APG entropy, concentration, and fourth-order defect across the three Matbench pilot datasets.

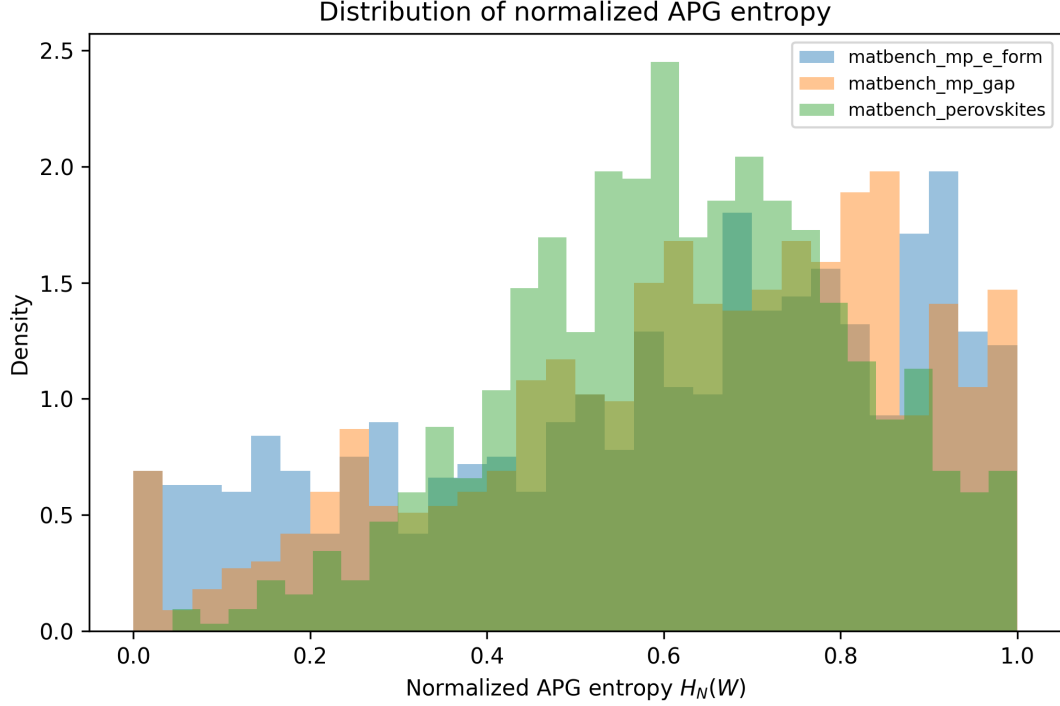


Figure 2: Distribution of normalized APG entropy $H_N(W)$ across datasets. Entropy shifts indicate different concentration regimes.

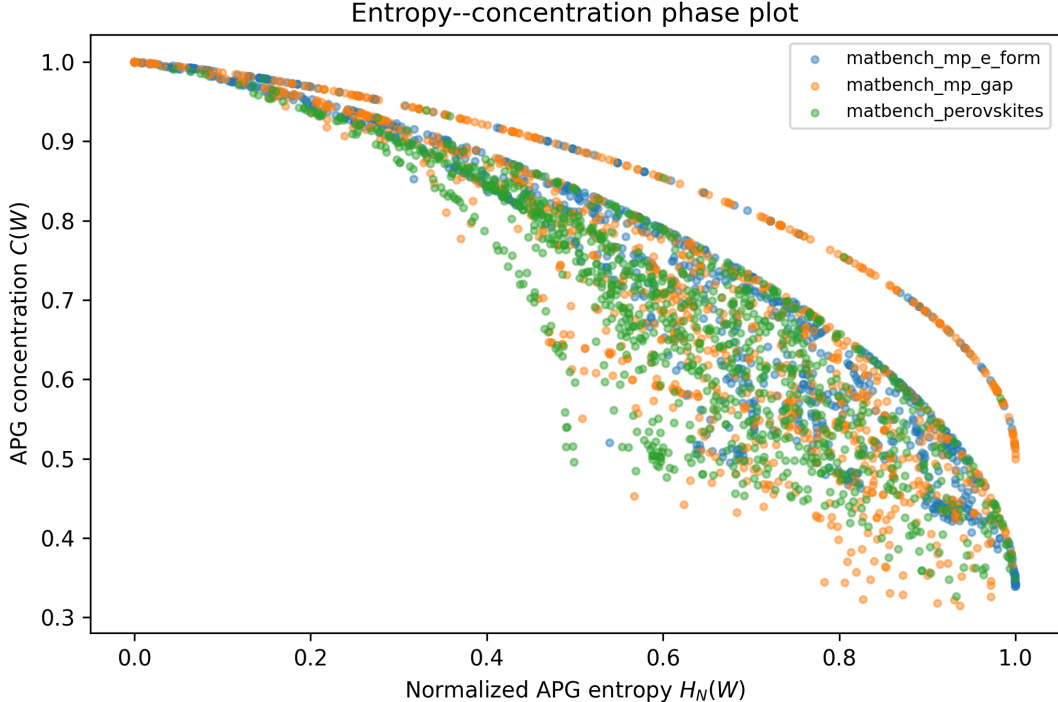


Figure 3: Entropy–concentration phase plot for the APG pilot. The inverse relation between $H_N(W)$ and $C(W)$ is expected but not redundant, because entropy uses the full mass distribution while concentration records only the maximum coordinate weight.

The joint behavior of entropy and concentration is shown in Figure 3, providing a geometric view of APG descriptor organization in the entropy–concentration plane.

Table 2 reports the one-vs-rest APG-OOD pilot. Treating `matbench_perovskites` as the in-distribution reference gives the strongest separability, with AUROC 0.9210. Treating `matbench_mp_e_form` as the reference gives AUROC 0.8426, which is the most relevant setting for future APG-based materials-discovery studies because formation energy is central to stability assessment and high-throughput screening. Treating `matbench_mp_gap` as the reference gives weaker AUROC 0.6172, suggesting that APG separation depends strongly on the chosen target domain and feature map.

The resulting APG-OOD performance is summarized in Figure 4, which compares AUROC values obtained when each dataset is treated as the in-distribution reference.

Table 2: Pilot one-vs-rest APG-OOD detection using Mahalanobis distance in APG descriptor space.

In-distribution Dataset	Metric	Value
perovskites	OOD dataset	all other datasets
perovskites	AUROC	0.9210
perovskites	Avg. precision	0.9555
perovskites	Mean ID score	8.0000
perovskites	Mean OOD score	42.9177
mp_gap	OOD dataset	all other datasets
mp_gap	AUROC	0.6172
mp_gap	Avg. precision	0.7820
mp_gap	Mean ID score	8.0000
mp_gap	Mean OOD score	11.2968
mp_e_form	OOD dataset	all other datasets
mp_e_form	AUROC	0.8426
mp_e_form	Avg. precision	0.8953
mp_e_form	Mean ID score	8.0000
mp_e_form	Mean OOD score	20.3297

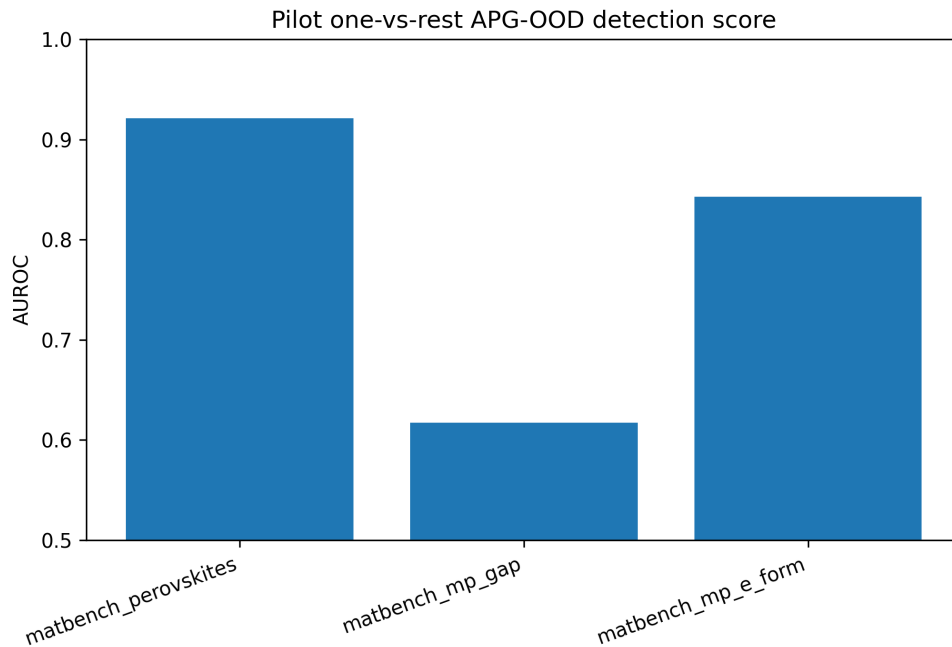


Figure 4: One-vs-rest APG-OD AUROC for the three reference choices.

7 Evaluation and Interpretation

The pilot study supports three principal observations. First, APG descriptors reveal measurable structural differences across the Matbench benchmark datasets, indicating that entropy, concentration, and exponent-deformation behavior vary systematically among distinct materials domains. Second, the strongest APG-based separation is observed for the perovskite benchmark. However, the formation-energy benchmark remains of particular practical interest because formation energy is closely associated with thermodynamic stability and materials screening. This finding is especially relevant to materials-discovery applications. Third, the present results do not indicate that APG functions as a superior materials-property predictor. Instead, the evidence supports APG as an interpretable descriptor framework capable of identifying structural differences between materials representations.

The observed behavior is consistent with the APG anomaly interpretation developed in earlier work, where entropy collapse and concentration growth were proposed as representation-dependent indicators of structural organization rather than universal laws (Akhtar, 2026d). Within materials informatics, this observation suggests that APG should be evaluated across multiple representation spaces, including composition descriptors, crystallographic descriptors, graph-derived features, and learned neural embeddings. Consequently, the absence of a particular APG pattern in one representation should not be interpreted as evidence against the framework itself. Rather, it may indicate that the selected representation does not expose the structural organization relevant to APG analysis.

8 Comparison With Existing Approaches

The proposed APG framework occupies a distinct position within the materials-informatics literature. Unlike Matbench leaderboard methods, APG is not designed to optimize predictive accuracy for band-gap prediction, formation-energy estimation, or related property-prediction tasks (Dunn et al., 2020). Similarly, APG does not perform crystal relaxation, energy prediction, or stability ranking, which are central objectives of Matbench Discovery (Riebesell et al., 2025). Instead, APG provides a compact structural characterization of materials representations that can be computed independently of a predictive model.

Compared with graph-based materials-learning architectures such as CGCNN, MEGNet, ALIGNN, M3GNet, and CHGNet, APG is not a neural architecture and does not learn interatomic interactions or atomistic potentials (Xie and Grossman, 2018; Chen et al., 2019; Choudhary and DeCost, 2021; Chen and Ong, 2022; Deng et al., 2023). Its contribution is descriptor-oriented. APG transforms a materials representation into interpretable entropy, concentration, and defect coordinates that summarize how representation mass is distributed across feature dimensions.

Relative to established out-of-distribution detection methods, APG is most closely related to feature-space approaches such as Mahalanobis scoring (Lee et al., 2018). However, APG differs fundamentally in the representation layer on which the analysis is performed. Rather than measuring distance directly within the original feature space, APG converts a representation into a probability structure based on squared-coordinate mass and then evaluates the organization of that structure. Deep ensembles and Bayesian-inspired approaches

estimate predictive uncertainty (Lakshminarayanan et al., 2017; Gal and Ghahramani, 2016), whereas APG quantifies structural unfamiliarity. Likewise, energy-based methods operate on classifier outputs or logits (Liu et al., 2020), while APG may be applied before prediction, after feature extraction, or within latent embedding spaces. APG should therefore be viewed as a complementary structural-analysis layer rather than a replacement for existing uncertainty-estimation or out-of-distribution detection approaches.

9 Implications for Materials Discovery

The results suggest that APG descriptors may provide a useful structural-monitoring framework for materials-discovery workflows. By mapping materials representations into an entropy–concentration–defect space, APG offers an interpretable mechanism for identifying samples that depart from the structural regimes represented in reference datasets. Such information may assist large-scale screening, model deployment under distribution shift, and candidate prioritization for additional computational validation.

The formation-energy benchmark is of particular interest because formation energy is closely linked to stability assessment and high-throughput discovery. The observed APG separability suggests that entropy and concentration descriptors may help characterize the boundary between familiar and unfamiliar materials representations while providing an additional layer of interpretability alongside predictive models.

10 Limitations

Several limitations should be acknowledged. First, the study uses pilot samples of 1,000 materials from each dataset rather than the complete benchmark datasets. Second, the feature construction relies on relatively simple composition and structure descriptors instead of modern graph-neural embeddings. Third, the out-of-distribution evaluation is based on cross-dataset one-versus-rest comparisons and does not include chemically controlled protocols such as leave-one-element, leave-one-system, leave-one-space-group, or leave-one-prototype experiments. Finally, the study focuses on structural characterization and distribution-shift analysis rather than predictive-property modeling. The results should therefore be interpreted as evidence of APG-based structural separability rather than as definitive evidence of deployment-ready out-of-distribution detection.

11 Future Directions

Several directions warrant further investigation. The most immediate extension is evaluation on the complete Matbench benchmark datasets. A second direction involves chemically meaningful out-of-distribution protocols, including leave-one-element, leave-one-system, leave-one-space-group, and leave-one-prototype experiments. Third, APG descriptors should be evaluated on learned representations generated by CGCNN, MEGNet, ALIGNN, M3GNet, and CHGNet. Fourth, systematic comparisons with established out-of-distribution methods, including Mahalanobis distance, Isolation Forest, Local Outlier Factor, deep ensembles,

and energy-based scores, would clarify the relative strengths and limitations of APG-based screening. Finally, APG may be integrated into end-to-end materials-discovery pipelines in which structurally unfamiliar candidates are flagged for additional first-principles verification.

12 Conclusion

This study introduced an APG framework for interpretable out-of-distribution analysis in materials discovery. Across three Matbench benchmark datasets, APG descriptors revealed distinguishable entropy–concentration–defect regimes under the proposed feature construction. The results demonstrate that APG can capture structural differences among materials domains and provide meaningful signals for distribution-shift analysis. Although APG is not intended to replace state-of-the-art materials-prediction models, it offers a complementary perspective focused on representation organization and structural unfamiliarity. These findings motivate further investigation using larger datasets, chemically controlled benchmarks, and modern graph-neural materials representations.

Data Availability

All datasets used in this study are publicly available through the Matbench benchmark (<https://matbench.materialsproject.org>). The APG feature-generation and evaluation code used in this study is available from the author upon reasonable request.

References

- Akhtar, M. A. K. (2026a). *Arithmetic Power Geometry 0: Unified foundations, axioms, core invariants, examples, and open problems*. Zenodo Preprint.
- Akhtar, M. A. K. (2026b). *Shannon–Rényi control of the Arithmetic Power Geometry closure defect*. Zenodo Preprint.
- Akhtar, M. A. K. (2026c). *Multi-variable Shannon–Rényi control laws in Arithmetic Power Geometry: Theory, empirical validation, and comparative analysis on cache-like information networks*. Zenodo Preprint.
- Akhtar, M. A. K. (2026d). *Entropy collapse and concentration growth as structural signatures of anomalous behavior: Evidence from financial fraud and blockchain transaction networks*. Zenodo Preprint.
- Amari, S. (2016). *Information geometry and its applications*. Springer.
- Cheng, J., Zhang, C., & Dong, L. (2021). A geometric-information-enhanced crystal graph network for predicting properties of materials. *Communications Materials*, 2, 92. <https://doi.org/10.1038/s43246-021-00194-3>

- Chen, C., Ye, W., Zuo, Y., Zheng, C., & Ong, S. P. (2019). Graph networks as a universal machine learning framework for molecules and crystals. *Chemistry of Materials*, 31(9), 3564–3572. <https://doi.org/10.1021/acs.chemmater.9b01294>
- Chen, C., & Ong, S. P. (2022). A universal graph deep learning interatomic potential for the periodic table. *Nature Computational Science*, 2, 718–728. <https://doi.org/10.1038/s43588-022-00349-3>
- Choudhary, K., & DeCost, B. (2021). Atomistic line graph neural network for improved materials property predictions. *npj Computational Materials*, 7, 185. <https://doi.org/10.1038/s41524-021-00650-1>
- Cover, T. M., & Thomas, J. A. (2006). *Elements of information theory* (2nd ed.). Wiley.
- Csiszár, I., & Körner, J. (2011). *Information theory: Coding theorems for discrete memoryless systems* (2nd ed.). Cambridge University Press.
- Deng, B., Zhong, P., Jun, K., Riebesell, J., Han, K., Bartel, C. J., & Ceder, G. (2023). CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence*, 5, 1031–1041. <https://doi.org/10.1038/s42256-023-00716-3>
- Dunn, A., Wang, Q., Ganose, A., Dopp, D., & Jain, A. (2020). Benchmarking materials property prediction methods: The Matbench test set and Automatminer reference algorithm. *npj Computational Materials*, 6, 138. <https://doi.org/10.1038/s41524-020-00406-3>
- Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning* (pp. 1050–1059).
- Gini, C. (1912). *Variabilità e mutabilità*. Tipografia di Paolo Cuppini.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.
- Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning* (pp. 1321–1330).
- Hendrycks, D., & Gimpel, K. (2017). A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *Proceedings of the International Conference on Learning Representations*.
- Jain, A., Ong, S. P., Hautier, G., Chen, W., Richards, W. D., Dacek, S., Cholia, S., Gunter, D., Skinner, D., Ceder, G., & Persson, K. A. (2013). Commentary: The Materials Project: A materials genome approach to accelerating materials innovation. *APL Materials*, 1(1), 011002. <https://doi.org/10.1063/1.4812323>
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems* (Vol. 25).

- Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems* (Vol. 30).
- Lee, K., Lee, K., Lee, H., & Shin, J. (2018). A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems* (Vol. 31).
- Li, K., Rubungo, A. N., Lei, X., Persaud, D., Choudhary, K., DeCost, B., Dieng, A. B., & Hattrick-Simpers, J. (2025). Probing out-of-distribution generalization in machine learning for materials. *Communications Materials*, 6, 9. <https://doi.org/10.1038/s43246-024-00731-w>
- Liang, S., Li, Y., & Srikant, R. (2018). Enhancing the reliability of out-of-distribution image detection in neural networks. In *Proceedings of the International Conference on Learning Representations*.
- Liu, W., Wang, X., Owens, J., & Li, Y. (2020). Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems* (Vol. 33, pp. 21464–21475).
- Marshall, A. W., Olkin, I., & Arnold, B. C. (2011). *Inequalities: Theory of majorization and its applications* (2nd ed.). Springer.
- Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., Dillon, J. V., Lakshminarayanan, B., & Snoek, J. (2019). Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems* (Vol. 32).
- Ren, J., Liu, P. J., Fertig, E., Snoek, J., Poplin, R., DePristo, M. A., Dillon, J. V., & Lakshminarayanan, B. (2019). Likelihood ratios for out-of-distribution detection. In *Advances in Neural Information Processing Systems* (Vol. 32).
- Rényi, A. (1961). On measures of entropy and information. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability* (Vol. 1, pp. 547–561). University of California Press.
- Riebesell, J., Goodall, R. E. A., Deng, B., Lee, A. A., Jain, A., & Persson, K. A. (2025). A framework to evaluate machine learning crystal stability predictions. *Nature Machine Intelligence*. <https://doi.org/10.1038/s42256-025-01055-1>
- Reiser, P., Neubert, M., Eberhard, A., Torresi, L., Zhou, C., Shao, C., Metni, H., van Hoesel, C., Schopmans, H., Sommer, T., & Friederich, P. (2022). Graph neural networks for materials science and chemistry. *Communications Materials*, 3, 93. <https://doi.org/10.1038/s43246-022-00315-6>
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423. <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>

- Sun, Y., Guo, C., & Li, Y. (2021). ReAct: Out-of-distribution detection with rectified activations. In *Advances in Neural Information Processing Systems* (Vol. 34).
- Trezza, G., & Chiavazzo, E. (2025). Classification-based detection and quantification of cross-domain data bias in materials discovery. *Journal of Chemical Information and Modeling*, 65(4), 1747–1761. <https://doi.org/10.1021/acs.jcim.4c01766>
- van Erven, T., & Harremoës, P. (2014). Rényi divergence and Kullback–Leibler divergence. *IEEE Transactions on Information Theory*, 60(7), 3797–3820. <https://doi.org/10.1109/TIT.2014.2320500>
- Xie, T., & Grossman, J. C. (2018). Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Physical Review Letters*, 120(14), 145301. <https://doi.org/10.1103/PhysRevLett.120.145301>
- Yang, J., Zhou, K., Li, Y., & Liu, Z. (2024). Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 132, 5635–5662. <https://doi.org/10.1007/s11263-024-02117-4>